#

Cloud-Based Random Forest Microservice for Predicting Building Energy Loads

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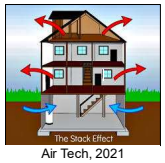
Introduction

- **Energy Efficiency:**
  - We utilized the Energy Efficiency Dataset [1] [3], from the UC Irivine Machine Learning Repository
- **Dataset Details:**
  - **768** simulated residential buildings [2]
  - Composed of **3.5 x 3.5 x 3.5-meter** cubes
  - All buildings have identical volume: **771.75 m^3**
- Each building is characterized by 8 input features**:**
  - **Relative Compactness, Surface Area, Wall Area, Roof Area, Overall Height, Orientation, Glazing Area, Glazing Area Distribution**
- The targets (outputs) that were predicted:
  - **Heating Load, Cooling Load**



Air Tech, 2021

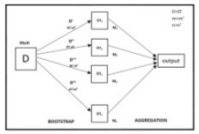
- **Random Forest Regressor (Multi-Output):**
  - Complexity of relationships within the dataset necessitates advanced ensemble modeling techniques to get robust results without overfitting
    - The ensemble method that builds many decision trees on random subsets of data and features
    - Each tree predicts independently, and the outputs are averaged for the final prediction
    - Introduced randomness reduces overfitting and improves generalization

Objective

We aim to predict building energy loads more accurately by deploying an optimized Random Forest model as a cloud-based microservice.

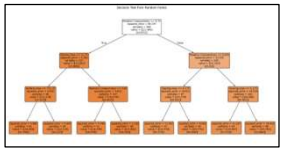
Methods

- **Training Setup:**
  - Model development and training performed in **Google Colab** [4] using **Scikit-Learn** libraries.
- **Multi-Output Random Forest Regressor Architecture:**
  - Number of Trees: 100
  - Maximum Tree Depth: 3
  - Minimum Samples to Split: 10
  - Minimum Samples per Leaf: 20
  - Maximum: Leaf Nodes: None
  - Random Seed: 42
  - Parallel Jobs: -1



geeksforgeeks, 2025

- **Multi-Output Random Forest Regressor Compilation**
  - **Preprocessing:**
    - **Standard Scaler:** Standardized features to mean ≈ 0 and standard deviation ≈ 1
    - **Train-Test Split:** 75% training / 25% testing
  - **Training & Optimization:**
    - **Grid Search (GridSearchCV):** Used for initial hyperparameter selection
    - **Manual Tuning:** Adjusted hyperparameters to improve generalization and reduce overfitting
- **Service Infrastructure:**
  - Docker Container:
    - Ubuntu 24.04.2
    - Python 3.11.12
    - Pandas 2.2.2
    - NumPy 2.0.2
    - Scikit-Learn 1.6.1
    - Matplotlib 3.10.0

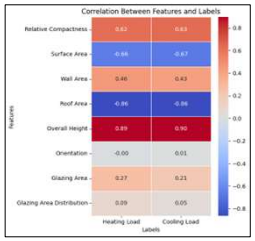


  - GitHub for version control
  - SwaggerHub for development of REST API
- **Evaluation Metrics:**
  - Used Scikit-Learn *accuracy, mean_squared_error, mean_absolute_error,* and *r2_score* libraries for performance metrics
  - Latency of endpoints tested using Python file
  - Four endpoints implemented for prediction, performance, and figure creation within cloud-service

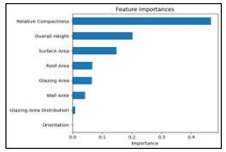
Results

Model Performance

- **Correlation Matrix**
  - Shows correlations between all the features and their targets, both positively and negatively
  - Relative Compactness and Overall Height exhibit strong positive correlations with load prediction
  - Roof Area and Surface Area show strong negative correlations with load prediction
  - Otherwise, features such as Orientation and Glazing Area have minimal impact on load prediction
- **Evaluation Metrics:**
  - **Accuracy Score**
    - 93.33%
  - **Mean Absolute Error**
    - 1.8772
  - **Mean Square Error**
    - 6.5556
  - **R^2 Score**
    - 0.9333



- **Feature Importances**
  - Displays the relative importance of each feature for predicting the target variables
  - Most Important Features
    - Relative Compactness
    - Overall Height
  - Least Important Features
    - Orientation
    - Glazing Area Distribution
  - Highlights which features have the greatest overall influence on the model's predictions



- **Service Performance**
  - The table shows the average latency for the three main endpoints with 10 request
  - Endpoints using uploaded data perform fastest

Average Latency with 10 request

Endpoint	March 13th, 3:40PM	March 14th, 12:10 AM
1. /Model_info	~0.0313s	~0.0040s
2. /Predict_only	~0.0387s	~0.0039s
3. /Prediction (With Upload)	~0.0066s	~0.0080s

Discussion and Future Work

- The Multi-Output Random Forest Regressor Model showed promising result in accurately quantitatively estimating the energy performance of residential buildings
- Demonstrates advanced machine learning approaches for forecasting building energy efficiency
- Limitations:
  - Simulated heating and cooling loads may not exactly match real-world building performance
  - Real world adaptability
- Future Work:
  - The inclusion of additional input variables (climate, material, etc.) and output variables (ventilation load, lighting load, etc.)
  - Implementing different hyperparameters
  - Use real world data
  - Analyze different building geometries

References

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